

First name:

Last name:

Student ID:

Name and Student ID must be filled in for this page to be graded.

Problem 1. Use **Spherical coordinates** to find the surface area of the cone $\phi = \frac{\pi}{6}$ with $0 \leq \rho \leq 2$.

For spherical coordinates:

$$h_1 = h_\rho = 1, \quad h_2 = h_\theta = \rho \sin \phi, \quad h_3 = h_\phi = \rho.$$

[5 marks]

Solution.

□

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Problem 2. Using double integrals and the **Cylindrical coordinates**, compute the surface area of a cylinder of radius 2, with $0 \leq z \leq 2$.

For cylindrical coordinates:

$$h_1 = h_r = 1, \quad h_2 = h_\theta = r, \quad h_3 = h_z = 1.$$

[5 marks]

Solution.

□